

ENGR 2405 — Exam 3 Formula Sheet

AC Circuits: Sadiku Chapters 9–12 and 14

Read this first. Murthy's provided sheet ("Useful Formulas", 4 pages) already gives you the items marked [given] below, so do *not* spend memorization effort on them. It does **not** give you the items marked [memorize], and several of those are unavoidable on an AC exam. The single most important gap: **the provided sheet has no impedance table and no phasor transform.** Its Passive Elements Table is entirely time-domain ($v = L di/dt$, $i = C dv/dt$); $j\omega L$ and $1/j\omega C$ appear nowhere. Neither does $V_{\text{rms}} = V_m/\sqrt{2}$, nor P_{max} , nor any three-phase power formula, nor Q and bandwidth.

[given] on the provided sheet

[memorize] not provided — memorize

A. The Gaps (memorize these first)

Everything here is absent from the provided sheet and is needed for Ch. 9–12, 14.

1. Phasor transform & impedances

$$V_m \cos(\omega t + \phi) \leftrightarrow V_m \angle \phi, \quad \sin \omega t = \cos(\omega t - 90^\circ)$$

$$\mathbf{Z}_R = R, \quad \mathbf{Z}_L = j\omega L, \quad \mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

2. RMS and the amplitude $\frac{1}{2}$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}, \quad P = \frac{1}{2} V_m I_m \cos \theta$$

The provided AC Power table is written *only* in RMS. If the problem gives you $10 \cos \omega t$, you must supply the $\frac{1}{2}$ yourself.

3. Maximum average power transfer

$$\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^*, \quad P_{\text{max}} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}} \text{ (ampl.)}$$

4. Power factor correction

$$Q_C = P(\tan \theta_1 - \tan \theta_2), \quad C = \frac{Q_C}{\omega V_{\text{rms}}^2}$$

5. Three-phase power (the sheet gives only the voltage relations)

$$P_T = \sqrt{3} V_L I_L \cos \theta, \quad S_T = \sqrt{3} V_L I_L$$

$$\mathbf{Z}_Y = \mathbf{Z}_\Delta / 3, \quad I_L = \sqrt{3} I_p (\Delta), \quad I_L = I_p (Y)$$

$$P_{1,2} = V_L I_L \cos(\theta \pm 30^\circ), \quad Q_T = \sqrt{3}(P_2 - P_1)$$

6. Resonance: Q and bandwidth

The sheet gives $\omega_0 = 1/\sqrt{LC}$ and α , but not Q or B . Useful bridge: $B = 2\alpha$ for both series and parallel.

$$Q = \frac{\omega_0}{B}, \quad B_{\text{ser}} = \frac{R}{L}, \quad B_{\text{par}} = \frac{1}{RC}$$

$$\omega_{1,2} \approx \omega_0 \mp B/2 \quad (Q \geq 10), \quad \omega_0 = \sqrt{\omega_1 \omega_2}$$

Practical parallel (coil R in series with L): $\omega_0 = \sqrt{1/LC - R^2/L^2}$.

7. Bode slopes and filter cutoffs

Pole -20 dB/dec, zero $+20$ dB/dec, $(j\omega)^{\pm N} : \pm 20N$ dB/dec.
 RC : $\omega_c = 1/RC$. RL : $\omega_c = R/L$. Half power = -3 dB.

8. Scaling

$$R' = K_m R, \quad L' = \frac{K_m}{K_f} L, \quad C' = \frac{C}{K_m K_f}$$

9. Complex arithmetic

Polar/rectangular conversion and the $\angle$ rules (§1). Nothing about complex numbers appears on the provided sheet.

10. Op-amp gains

The sheet gives only the summer and difference amplifier. You must know $\mathbf{H} = -\mathbf{Z}_f/\mathbf{Z}_i$ (inverting) and $1 + \mathbf{Z}_f/\mathbf{Z}_i$ (non-inverting).

Already on the provided sheet — skip these.

Ohm's law; KVL; KCL; series and parallel R ; Δ -Y resistance transformation (full general form); Norton relation $I_N = V_{\text{Th}}/R_N$ with $R_N = R_{\text{Th}}$; 2×2 and 3×3 determinants for Cramer's rule; the complete complex/apparent/real/reactive power and pf definitions; the V_{an}/V_{ab} phase-to-line relations for positive sequence; the transfer-function standard form; $\omega_0 = 1/\sqrt{LC}$; the passive element table (series/parallel L and C , DC behavior); and all of first- and second-order transient response. Pages 3–4 are Laplace tables, which are outside this exam's scope.

0. Complex Number Arithmetic

$$z = x + jy = r \angle \phi, \quad r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}$$

$$x = r \cos \phi, \quad y = r \sin \phi$$

Add / subtract in rectangular; **multiply / divide** in polar:

$$r_1 \angle \phi_1 \cdot r_2 \angle \phi_2 = r_1 r_2 \angle (\phi_1 + \phi_2)$$

$$\frac{r_1 \angle \phi_1}{r_2 \angle \phi_2} = \frac{r_1}{r_2} \angle (\phi_1 - \phi_2)$$

$$z^* = x - jy = r \angle -\phi, \quad \frac{1}{j} = -j, \quad j^2 = -1$$

$$\sqrt{z} = \sqrt{r} \angle \frac{\phi}{2}, \quad j = 1 \angle 90^\circ$$

Calculator quadrant. $\tan^{-1}(y/x)$ returns only $\pm 90^\circ$. If $x < 0$, add or subtract 180° . Always sketch the point before trusting the angle.

1. Fundamentals (carry over to AC)

Ohm: $v = iR \Rightarrow \mathbf{V} = \mathbf{I}Z$

KCL: $\sum i_{\text{in}} = \sum i_{\text{out}}$ at any node or supernode

KVL: $\sum v = 0$ around any closed loop

Series / parallel:

$$\mathbf{Z}_{\text{ser}} = \sum \mathbf{Z}_k, \quad \frac{1}{\mathbf{Z}_{\text{par}}} = \sum \frac{1}{\mathbf{Z}_k}$$

$$\mathbf{Z}_1 \parallel \mathbf{Z}_2 = \frac{\mathbf{Z}_1 \mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$$

Voltage divider: $\mathbf{V}_k = \mathbf{V}_s \frac{\mathbf{Z}_k}{\sum \mathbf{Z}}$

Current divider (2 branches): $\mathbf{I}_1 = \mathbf{I}_s \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}$

Source transformation: v_s in series with $\mathbf{Z} \Leftrightarrow v_s/\mathbf{Z}$ in parallel with $\mathbf{Z}$

Thévenin / Norton:

$$\mathbf{I}_N = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}}}, \quad \mathbf{Z}_N = \mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_{oc}}{\mathbf{I}_{sc}}$$

Dependent sources present $\Rightarrow$ use $\mathbf{V}_{oc}/\mathbf{I}_{sc}$ or a $1 \angle 0^\circ$ V test source with independent sources killed.

Kill a source: voltage source $\rightarrow$ short, current source $\rightarrow$ open. Never kill a dependent source.

Superposition: valid for v and i , *not* for power.

2. Sinusoids and Phasors (Ch. 9)

$$v(t) = V_m \cos(\omega t + \phi), \quad \omega = 2\pi f = \frac{2\pi}{T}$$

Convert to cosine first:

$$\sin(\omega t) = \cos(\omega t - 90^\circ)$$

$$-\cos(\omega t) = \cos(\omega t \pm 180^\circ)$$

$$-\sin(\omega t) = \cos(\omega t + 90^\circ)$$

Time domain	Phasor
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
dv/dt	$j\omega \mathbf{V}$
$\int v dt$	$\mathbf{V}/j\omega$

Impedance and admittance:

	$\mathbf{Z}$	$\mathbf{Y}$
R	R	$1/R$
L	$j\omega L$	$1/(j\omega L)$
C	$-j/(\omega C)$	$j\omega C$

$$\mathbf{Z} = R + jX, \quad \mathbf{Y} = \frac{1}{\mathbf{Z}} = G + jB$$

$X > 0$ inductive, $X < 0$ capacitive. Note $G \neq 1/R$ in general.

Limits: at $\omega = 0$ (DC), C is open and L is short. At $\omega \rightarrow \infty$, C is short and L is open.

ELI the ICE man. In L , voltage (E) leads current (I) by 90° . In C , current leads voltage by 90° .

3. AC Analysis (Ch. 10)

All DC methods apply with $R \rightarrow \mathbf{Z}$.

Mesh matrix (all currents clockwise, independent voltage sources only) is symmetric:

$$\begin{bmatrix} \sum \mathbf{Z}_{11} & -\mathbf{Z}_{12} \\ -\mathbf{Z}_{12} & \sum \mathbf{Z}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

Cramer's rule (2×2):

$$\Delta = ad - bc, \quad x_1 = \frac{e_1 d - b e_2}{\Delta}, \quad x_2 = \frac{a e_2 - e_1 c}{\Delta}$$

Supernode: KCL on the enclosing surface, plus $\mathbf{V}_1 - \mathbf{V}_2 = \mathbf{V}_s$.

Supermesh: KVL around the outside, plus $\mathbf{I}_2 - \mathbf{I}_1 = \mathbf{I}_s$.

Op amp (ideal): $\mathbf{V}_+ = \mathbf{V}_-$, no input current.

$$\text{inverting } \mathbf{H} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i}, \quad \text{non-inv } \mathbf{H} = 1 + \frac{\mathbf{Z}_f}{\mathbf{Z}_i}$$

Different source frequencies $\Rightarrow$ superposition is mandatory. Solve each ω in its own phasor circuit, then add the *time-domain* results. Phasors at different frequencies can never be added.

4. AC Power (Ch. 11)

With $\mathbf{V} = V_m \angle \theta_v$, $\mathbf{I} = I_m \angle \theta_i$, $\theta = \theta_v - \theta_i = \angle \mathbf{Z}$:

$$p(t) = \frac{1}{2} V_m I_m \cos \theta + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Amplitude phasors	RMS phasors
$P = \frac{1}{2} V_m I_m \cos \theta$	$P = V_{\text{rms}} I_{\text{rms}} \cos \theta$
$Q = \frac{1}{2} V_m I_m \sin \theta$	$Q = V_{\text{rms}} I_{\text{rms}} \sin \theta$
$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^*$	$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \text{ (sinusoids only)}, \quad V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

$$\mathbf{S} = P + jQ = S \angle \theta, \quad S = |\mathbf{S}| = V_{\text{rms}} I_{\text{rms}}$$

$$\mathbf{S} = I_{\text{rms}}^2 \mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*}, \quad \text{pf} = \cos \theta = \frac{P}{S}$$

By element: R absorbs only P ; L absorbs $+Q$ only; C absorbs $-Q$ only.

Sign of θ : $\theta > 0$ inductive / lagging pf / $\mathbf{I}$ lags $\mathbf{V}$. $\theta < 0$ capacitive / leading pf.

Conservation: P and Q each conserved; add loads as $P + jQ$. S is *not* conserved.

Max average power transfer:

$$\mathbf{Z}_L = \mathbf{Z}_{Th}^*, \quad P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}} \text{ (ampl.)} = \frac{|\mathbf{V}_{Th}|^2}{4R_{Th}} \text{ (rms)}$$

If $\mathbf{Z}_L$ must be purely resistive: $R_L = |\mathbf{Z}_{Th}|$.

Power factor correction (shunt C , lagging $\rightarrow$ less lagging):

$$Q_C = P(\tan \theta_1 - \tan \theta_2), \quad C = \frac{Q_C}{\omega V_{\text{rms}}^2}$$

5. Three-Phase (Ch. 12)

Positive (abc) sequence: $\mathbf{V}_{an} = V_p \angle 0^\circ$, $\mathbf{V}_{bn} = V_p \angle -120^\circ$, $\mathbf{V}_{cn} = V_p \angle +120^\circ$. Negative (acb) reverses the last two. Balanced $\Rightarrow$ the three phasors sum to zero.

	Y	Δ
Voltage	$V_L = \sqrt{3}V_p$ $\mathbf{V}_{ab} = \sqrt{3}\mathbf{V}_{an} \angle 30^\circ$	$V_L = V_p$
Current	$I_L = I_p$	$I_L = \sqrt{3}I_p$ $\mathbf{I}_a = \sqrt{3}\mathbf{I}_{AB} \angle -30^\circ$

Same formulas for both connections:

$$P_T = \sqrt{3} V_L I_L \cos \theta, \quad Q_T = \sqrt{3} V_L I_L \sin \theta$$

$$S_T = \sqrt{3} V_L I_L = 3V_p I_p, \quad \mathbf{S}_T = P_T + jQ_T$$

Per-phase form: $P_T = 3I_p^2 R_p$.

Conversions: $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$; Δ source to Y: $V_p = V_L/\sqrt{3}$ with a -30° shift.

Balanced systems: neutral current is zero, instantaneous total power is constant ($p_T = P_T$, no 2ω ripple).

Two-wattmeter method (meters in lines a , b ; coils referenced to c):

$$P_1 = V_L I_L \cos(\theta + 30^\circ), \quad P_2 = V_L I_L \cos(\theta - 30^\circ)$$

$$P_T = P_1 + P_2, \quad Q_T = \sqrt{3}(P_2 - P_1)$$

$$\theta = \tan^{-1} \left[\sqrt{3} \frac{P_2 - P_1}{P_2 + P_1} \right]$$

$\theta = 0 \Rightarrow$ equal readings; $\theta = 60^\circ \Rightarrow$ one reads zero; $\theta > 60^\circ \Rightarrow$ one reads negative.

Unbalanced four-wire Y–Y: treat each phase alone; then $\mathbf{I}_{nN} = -(\mathbf{I}_{aA} + \mathbf{I}_{bB} + \mathbf{I}_{cC}) \neq 0$. The $\sqrt{3}$ formulas do not apply — sum the three phase powers.

θ is the angle of the **per-phase load impedance**, never the angle of $\mathbf{I}_L$ or $\mathbf{V}_L$.

6. Frequency Response (Ch. 14)

$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)} = H \angle \phi$$

Decibels:

$$G_{\text{dB}} = 20 \log_{10} |\mathbf{H}| \text{ (voltage / current)}$$

$$G_{\text{dB}} = 10 \log_{10} (P_2/P_1) \text{ (power)}$$

Half power = -3 dB , i.e. $|\mathbf{H}| = H_{\max}/\sqrt{2} = 0.707 H_{\max}$.

Bode straight-line rules. First put $\mathbf{H}$ in standard form so every factor reads $(1 + j\omega/\omega_k)$.

Factor	Slope	Phase
K	0 at $20 \log K $	0°
$(j\omega)^N$	$+20N \text{ dB/dec}$	$+90N^\circ$
$(j\omega)^{-N}$	$-20N \text{ dB/dec}$	$-90N^\circ$
zero $(1 + j\omega/z)$	$+20$ after z	$0 \rightarrow +90^\circ$
pole $(1 + j\omega/p)^{-1}$	-20 after p	$0 \rightarrow -90^\circ$

Phase transitions run from $\omega_k/10$ to $10\omega_k$ at $\pm 45^\circ/\text{dec}$. Quadratic factors double every number ($\pm 40 \text{ dB/dec}$, $\pm 180^\circ$).

Resonance.

	Series RLC	Parallel RLC
ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Q	$\frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L} = \omega_0 RC$
B	$\frac{R}{L}$	$\frac{1}{RC}$

Always: $B = \omega_2 - \omega_1$, $Q = \omega_0/B$, $\omega_0 = \sqrt{\omega_1 \omega_2}$.

Exact half-power frequencies (series):

$$\omega_{1,2} = \mp \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

High Q ($Q \geq 10$): $\omega_{1,2} \approx \omega_0 \mp B/2$.

At series resonance: $\mathbf{Z} = R$ (minimum), $\mathbf{I}$ maximum and in phase with $\mathbf{V}_s$, pf = 1, and

$$|\mathbf{V}_L| = |\mathbf{V}_C| = Q |\mathbf{V}_s|$$

Practical parallel resonance — coil resistance R in series with L , all in parallel with C :

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

Do *not* use $1/\sqrt{LC}$ here.

Passive filters.

Filter	H and cutoff
<i>RC</i> low-pass	$\frac{1}{1 + j\omega RC}, \omega_c = \frac{1}{RC}$
<i>RC</i> high-pass	$\frac{j\omega RC}{1 + j\omega RC}, \omega_c = \frac{1}{RC}$
<i>RL</i> low-pass	$\omega_c = R/L$
<i>RL</i> high-pass	$\omega_c = R/L$
Band-pass	series <i>RLC</i> , V_o across <i>R</i>
Band-stop	series <i>RLC</i> , V_o across <i>LC</i>

Band-pass and band-stop: $\omega_0 = 1/\sqrt{LC}$, $B = R/L$.

Active filters (first order, inverting).

$$\text{low-pass: } \mathbf{H} = -\frac{R_f}{R_i} \cdot \frac{1}{1 + j\omega R_f C_f}, \omega_c = \frac{1}{R_f C_f}$$

$$\text{high-pass: } \mathbf{H} = -\frac{j\omega R_f C_i}{1 + j\omega R_i C_i}, \omega_c = \frac{1}{R_i C_i}$$

Band-pass = low-pass stage cascaded with high-pass stage;
overall gain is the product of the stage gains.

Scaling. Magnitude factor K_m , frequency factor K_f :

$$R' = K_m R, \quad L' = \frac{K_m}{K_f} L, \quad C' = \frac{C}{K_m K_f}$$

ω_0 and B scale by K_f ; Q is unchanged by either.

7. Constants and Conversions

$$\sqrt{3} = 1.7321, \quad \frac{1}{\sqrt{3}} = 0.5774, \quad \sqrt{2} = 1.4142$$

$$\frac{1}{\sqrt{2}} = 0.7071, \quad \tan 30^\circ = 0.5774, \quad \tan 60^\circ = 1.7321$$

Common power-factor angles:

pf	0.6	0.707	0.8	0.9	0.95
θ	53.13°	45°	36.87°	25.84°	18.19°
$\tan \theta$	1.333	1.000	0.750	0.484	0.329

US line frequency $f = 60 \text{ Hz}$, so $\omega = 2\pi(60) = 376.99 \text{ rad s}^{-1}$.

Before you submit: Did every sine become a cosine? Is the $\frac{1}{2}$ in the right place? Is θ the impedance angle? Does the final answer carry ωt ?